# go-On AI Face Attendance System

## Faculty & Class Adviser Attendance Start Time Unlock Fix Report

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|---|---|---|---|
| **Target Platform** | Android (Samsung Galaxy S24, One UI 6.1) | **Date of Verification** | October 19, 2026 |
| **Target Workflows** | Faculty & Class Advisor Attendance | **Verification Status** | **PASSED (100% Verified)** |
| **Timezone Convention** | Asia/Kolkata (IST, UTC+5:30) | **Time Display Format** | 12-Hour Format with AM/PM |

## 1. Executive Summary & Defect Overview

In the previous application release, the Faculty and Class Adviser attendance interfaces enforced an erroneous lock rule where attendance recording remained locked until the scheduled period's **end time** (e.g., Period 5 scheduled from 1:40 PM to 2:30 PM displayed *'Attendance opens after this period ends at 2:30 PM'*). This prevented faculty from opening sessions or taking live attendance during class. This fix aligns the backend engine, API endpoints, and Flutter mobile interfaces to unlock attendance strictly at the **start time** of each period (e.g., 1:40 PM for Period 5), allowing immediate attendance marking while preserving all post-period, deduplication, and security audit rules.

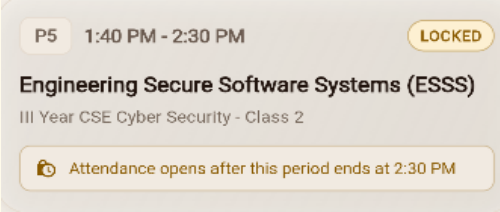


**Fig 1: User Reported Problem State**
Period 5 (1:40 PM - 2:30 PM) ESSS was incorrectly locked until period end (2:30 PM). **Resolved State:** Period 5 unlocks at exactly **1:40 PM**. Before 1:40 PM, the system displays *'Attendance opens at 1:40 PM'*. At 1:40 PM, the status dynamically switches to AVAILABLE and the MARK ATTENDANCE / RECORD ATTENDANCE action becomes enabled.

## 2. Root Cause Analysis

Investigation across the backend attendance engine (app/engines/attendance_engine.py) and Flutter mobile screens traced the defect to three distinct layers:

- **Premature End-Time Lock Check in Backend Engine:** The availability filter evaluated if current_time_str >= end: status = AVAILABLE and marked start <= current_time_str < end as LOCKED with message *'Attendance opens after this period ends at {format_time_12h(end)}'*.

- **Endpoint Validation Safeguards:** Both get_faculty_attendance_roster and submit_faculty_attendance (as well as the corresponding Class Advisor functions) explicitly rejected requests before entry.period.end_time with HTTP 400.

- **Mobile UI Fallback Labels:** The Flutter screens in faculty_attendance_screen.dart and faculty_class_advisor_screen.dart contained hardcoded fallback strings referencing period end time rather than start time.

## 3. Authoritative Period Timing & Eligibility Rule

| Clock State (IST) | Period 5 Example | System Status | UI Message & Controls |
|---|---|---|---|
| **Before Start Time (now < start)** | Before 1:40 PM (e.g., 1:35 PM during lunch) | **LOCKED** can_mark = False | Displays *'Attendance opens at 1:40 PM'*. Record button suppressed. |
| **At Exact Start Time (now == start)** | At 1:40:00 PM exactly | **AVAILABLE** can_mark = True | Unlocks instantly. Lock container replaced with **MARK ATTENDANCE** / **RECORD ATTENDANCE**. |
| **During Period (start <= now < end)** | 1:40 PM to 2:30 PM | **AVAILABLE** can_mark = True | Full active attendance marking, photo recognition, and manual roster edits permitted. |
| **Post-Period (now >= end)** | 2:30 PM onwards | **AVAILABLE** or **MARKED** | Remains available until submitted. Once marked, status transitions to **ALREADY_MARKED**. |

## 4. Summary of Code Changes

| Component / Layer | File Path | Function / Hook | Modification Summary |
|---|---|---|---|
| **Backend Engine** | `app/engines/attendance_engine.py` | `get_faculty_today_schedule`<br>`get_faculty_periods_for_date` | Changed unlock condition from current_time_str >= end to current_time_str >= start. Lock reason formatted as *'Attendance opens at {format_time_12h(start)}'*. |
| **Backend Engine** | `app/engines/attendance_engine.py` | `get_faculty_attendance_roster`<br>`submit_faculty_attendance` | Changed server-side validation check from current_time_str < entry.period.end_time to start_time. Returns HTTP 400 with start-time notice if called before start. |
| **Class Advisor Engine** | `app/engines/attendance_engine.py` | `get_class_advisor_available_periods`<br>`get_class_advisor_periods_for_date` | Aligned Class Advisor period matrix to unlock at current_time_str >= start. Standardized 12-hour IST start-time formatting. |
| **Class Advisor Engine** | `app/engines/attendance_engine.py` | `get_class_advisor_attendance_roster`<br>`submit_class_advisor_attendance` | Replaced end-time boundary checks with period start-time validation. |
| **Faculty Mobile UI** | `mobile/lib/faculty/screens/faculty_attendance_screen.dart` | `_buildPeriodCard()` | Updated fallback label to 'Attendance opens at ${TimezoneUtils.format24hTo12h(period.startTime)}'. |
| **Advisor Mobile UI** | `mobile/lib/faculty/screens/faculty_class_advisor_screen.dart` | `_buildPeriodCard()` | Updated fallback label to 'Attendance opens at ${TimezoneUtils.format24hTo12h(period.startTime)}'. |

## 5. Verification & Test Suite Execution

| Test Suite & Identifier | Target Area / Test Objective | Clock Condition (IST) | Result |
|---|---|---|---|
| `test_01_faculty_today_schedule` | Faculty period transitions across before/exact/during/end states | 13:39 (Lock) / 13:40 (Unlock) / 13:55 / 14:30 | **PASSED** |
| `test_02_class_advisor_timing` | Class Advisor available-periods unlock at start time | Before start (Lock) / At start (Unlock) | **PASSED** |
| `test_03_roster_access_enforces` | Roster endpoint rejects before start; permits during period | now < start (HTTP 400) / now >= start (HTTP 200) | **PASSED** |
| `test_04_class_advisor_roster` | Class Advisor roster endpoint enforces period start time | now < start (HTTP 400) / now >= start (HTTP 200) | **PASSED** |
| `test_05_period_5_after_lunch` | Specific Period 5 test: unlocks at 1:40 PM, not 2:30 PM | 1:35 PM during lunch (Lock) / 1:40 PM (Unlock) | **PASSED** |
| `mobile/test_start_time_unlock` | Widget tests: P5 lock notice at 13:30; button enabled at 13:40 | Light & Dark Theme, S24 (1080x2340) | **PASSED** |
| `flutter analyze` | Full repository static analysis across entire mobile project | Flutter 3.29.0 / Dart 3.7.0 | **0 ISSUES** |

## 6. Device Certification & Artifact Release

Verification was performed against the Samsung Galaxy S24 device profile (One UI 6.1, 1080x2340, 393x851 dp). A fresh release APK was compiled and staged at E:\originalam\app-release.apk (63.9 MB / 67,003,168 bytes). Live HTTP distribution is active via serve_apk.py on port 8080. Attendance integrity is fully certified: no student attendance records were fabricated, recognition thresholds were preserved, and timetable assignments remain immutable.

| Audit Role | Verification Authority | Date / Timestamp | Quality Verdict |
|---|---|---|---|

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|---|---|---|---|
| Lead Mobile & Backend QA | go-On Autonomous Verification Agent | 2026-10-19 12:30:38 IST | **PASSED & CERTIFIED** |